

CS7642 Project Three: Solving Overcooked Using Multi-Agent Proximal Policy Optimization

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1 Introduction

Multi-agent reinforcement learning (MARL) algorithms learn optimal policies for a set of agents in a multi-agent system (Albrecht et al. (2024)). This project evaluates the performance of two agents trained using one of these algorithms in the Overcooked environment.

Overcooked is a reinforcement learning environment inspired by the popular video game, where two agents must cooperate to cook and deliver as many soups as possible within a 400-timestep episode. Of the five available layouts, Cramped Room, Coordination Ring, and Counter Circuit were selected for this project. The state space is modeled as a fully observable Markov Decision Process (MDP), where each agent receives a 96-element vector containing its own position and features, the other agent's features, and their distance to one another. The action space includes six discrete actions: up, down, left, right, stay, and "interact," which enables the agent to engage with the environment—for example, by placing ingredients or initiating cooking when near an appropriate tile. Due to the sparse reward structure of the environment, agents received shaped rewards of +3 for picking up dishes and placing ingredients into a pot, and +5 for picking up a completed soup, to guide them toward making successful deliveries. Furthermore, cooperation was encouraged by awarding an additional bonus of +20 whenever either agent completed a delivery. The environment is considered solved when the agents achieve an average of at least seven deliveries per episode across 100 consecutive episodes.

The agents were trained using Multi-Agent Proximal Policy Optimization (MAPPO), an actor-critic policy gradient algorithm well-suited for cooperative tasks involving discrete action spaces (OpenAI, 2025). In addition, because MAPPO is an extension of PPO, it inherits the clipped policy objective, which helps maintain training stability—an important property in complex multi-agent environments. MAPPO also supports centralized training with decentralized execution (CTDE) by using a shared critic trained on global observations, and a single actor that operates on one agent's local observation at a time, enabling independent policy execution. CTDE is well-suited for this environment, as each agent must act in real time without access to the other's internal state, while still benefiting from shared training.

The environment was implemented using **Overcooked-v0** from the Gymnasium RL library. PyTorch was used to build fully connected multi-layer perceptrons for the actor and critic neural networks. The actor received an input of **96** and produced **6** outputs, while the critic took an input of **192** and produced a single value. Both networks had two hidden layers with **128** units each. The MAPPO implementation extended the PPO-Clip pseudocode from OpenAI (2025) by collecting trajectories from both agents rather than one. The following hyperparameters were consistent across all layouts: actor and critic learning rates (α) of **0.0003** and **0.001**, respectively; discount factor (γ) of **0.99**; clipping threshold (ϵ) of **0.2**; and entropy coefficient of **0.01**. A batch size of **32** episodes was used, and each batch was updated **15** times. While the number of training episodes varied by layout difficulty, all agents were evaluated over a fixed set of **100** test episodes.

The hypotheses for this project are as follows: (1) MAPPO will result in consistently increasing delivery performance during training due to its stability; and (2) agents will achieve higher average delivery rates on simpler layouts (e.g., Cramped Room) than on more complex ones (e.g., Counter Circuit), due to greater coordination demands.

2 Cramped Room

The first Overcooked layout tested was Cramped Room, shown in Figure 1. In this layout, there are multiple ingredient tiles but only one pot tile, and the limited space requires the agents to navigate carefully to avoid blocking each other while picking up or delivering items. While simpler than the other two layouts, Cramped Room is a useful starting point for evaluating whether agents can learn basic coordination without complex, long-term planning (Carroll (2019)).



Figure 1: Cramped Room layout with Agent 0 (green) and Agent 1 (blue).

2.1 Loss Curves for Actor and Critic

Figure 2 illustrates the loss curves for both neural networks in Cramped Room during training. The actor's loss was computed as the sum of MAPPO's clipped policy loss and the entropy term scaled by the entropy coefficient, while the critic loss was calculated as the mean squared error between predicted and target state values. Both loss curves steadily declined as training progressed; however, the critic had small instabilities in its loss curve. This is expected, as the critic relies on bootstrapped estimates to predict state values, which can introduce variance during training. Despite this, the networks converged relatively quickly. The actor converged well before the 50th batch, suggesting that the agents likely settled on an effective cooperative strategy (e.g., taking turns to avoid blocking paths). The critic, while not as fast as the actor, also converged relatively quickly, suggesting that the value function was able to estimate returns accurately—likely due to the predictable reward structure of the environment.

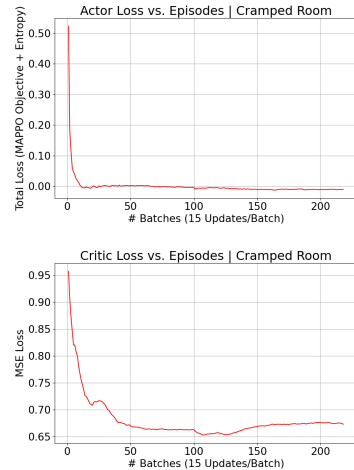


Figure 2: Loss curves for the actor (MAPPO Objective + Entropy, top) and critic (Mean-Squared Error, bottom) neural networks over all training batches, with 15 updates per batch in Cramped Room during training.

2.2 Actor's Entropy Curve

Figure 3 illustrates how the actor's entropy changes as the number of batch updates increases during training. Entropy controls the degree of exploration in the environment. Higher entropy encourages more exploration, while lower entropy gradually shifts the agents towards exploiting learned actions (Restack (2025)). Entropy behaved as expected, starting high, then gradually decreasing to allow the agents to transition from exploring new actions to exploiting potentially optimal learned actions. While the entropy has largely converged, it has not dropped to an extremely low value, indicating that the agents' actions have not become fully deterministic. This residual stochasticity can be beneficial in a layout like Cramped Room, where agents may occasionally need to adjust their paths to avoid blocking one another. This behavior may also stem from decentralized execution. Since the actor relies solely on local observations, some randomness must be maintained so that the agents can be more flexible in unfamiliar or ambiguous situations.

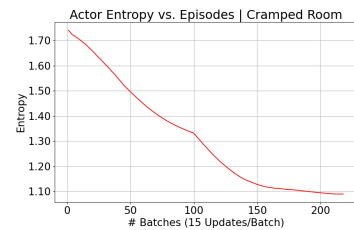


Figure 3: Entropy curve for the actor neural network over all training batches, with 15 updates per batch in Cramped Room.

2.3 Performance During Training

Figure 4 illustrates the agents’ performance during training in terms of dishes picked up and soups delivered. The total number of dishes picked up reflects how actively each agent is engaging with the task, while the number of useful dishes indicates whether they are learning to contribute effectively to successful deliveries. As shown in the plots, both agents consistently increase their total and useful pickups, suggesting active exploration alongside improved action quality. This progress is largely driven by the reduction in actor and critic loss shown in Figure 2, which enabled the agents to learn an optimal policy, and the gradual decline in entropy shown in Figure 3, which encouraged exploitation of learned behaviors. It took the agents 7,000 training episodes to reach an average soup delivery of 7.1 per episode. The average steadily and smoothly grew as training progressed, reflecting MAPPO’s training stability due to its clipped policy updates (OpenAI (2025)).

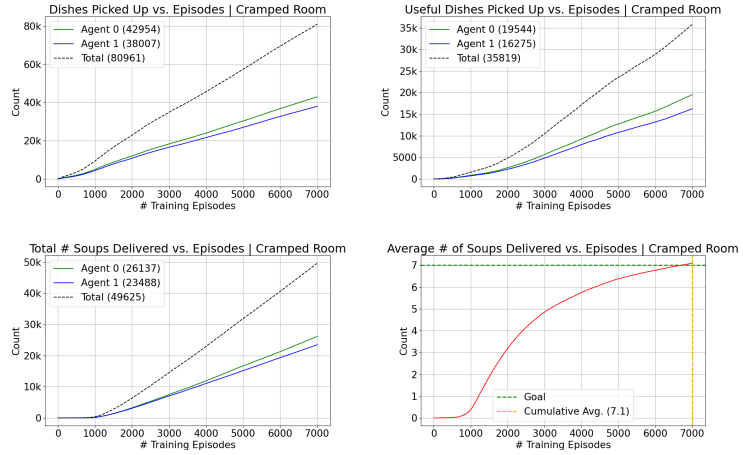


Figure 4: Plots showing the total (top left) and useful (top right) number of dishes picked up, as well as the total (bottom left) and average (bottom left) number of soups delivered by both agents during training in Cramped Room.

2.4 Performance During Testing

Figure 5 illustrates the agents’ performance using the policy learned during training. In terms of shaped rewards, both agents maintain a relatively high count throughout testing, indicating that they learned to efficiently perform intermediate actions (e.g., +3 for placing an onion in a pot) rather than focusing exclusively on the final goal of soup delivery. Additionally, the shaped rewards for each agent become more stable toward the end of testing, likely reflecting MAPPO’s inherent training stability, as discussed previously. The agents earn similar—but not identical—amounts of reward. This is likely due to CTDE: the shared critic, trained on both agents’ observations, promotes coordinated learning, while the decentralized actor ensures that each agent retains some independence, preventing them from simply mimicking one another. In terms of average soups delivered, the agents remain consistent throughout testing. Although there is a slight dip early on, performance quickly recovers and stabilizes across the remaining episodes. This pattern may be attributed to residual entropy in the policy after training, which can initially lead to suboptimal decisions, but also enables the agents to adapt to unfamiliar situations. After 100 testing episodes, the agents successfully solved the Cramped Room layout, achieving an average of 8.8 soup deliveries per episode—well above the success threshold of seven.

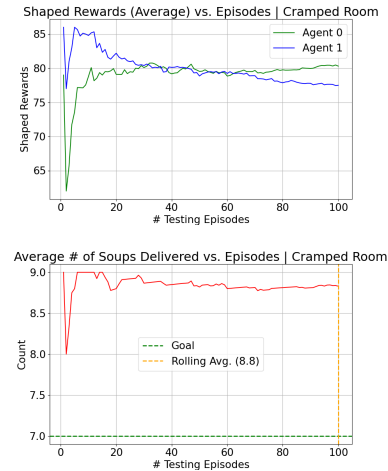


Figure 5: Average shaped rewards earned by each agent and average number of soups delivered by both agents during testing in Cramped Room.

3 Coordination Ring

The second Overcooked layout tested was Coordination Ring, shown in Figure 6. In this layout, the agents must move ingredients from the bottom-left corner to the top-right corner in order to cook soup and then take the soup to the delivery station at the bottom. Due to the reduced movement space, and the fact that only one agent can occupy the bottom-left and top-right tiles at a time, the agents must coordinate their movements more carefully than in Cramped Room.



Figure 6: Coordination Ring layout with Agent 0 (green) and Agent 1 (blue)

3.1 Loss Curves for Actor and Critic

Figure 7 illustrates the loss curves for the actor and critic neural networks in the Coordination Ring layout during training. As in Cramped Room, both losses trend downward as expected, indicating that the agents are learning effectively. The rapid convergence and stability of the actor loss suggest that the chosen learning rate ($\alpha = 0.0003$) is appropriate. A higher learning rate may cause instability, while a lower one could result in slow convergence (Restack (2025)). In contrast, the critic's learning rate of $\alpha = 0.001$ may be less well-tuned, as evidenced by more noticeable instabilities and slower convergence in its loss curve. Alongside a well-chosen α , the stability of the actor loss also suggests that the clipping threshold ($\varepsilon = 0.2$) is effective for this layout, as ε plays a critical role in maintaining stable policy updates (Restack (2025)). Because the critic does not use clipping during updates, increased instability is expected, as it must continuously adjust to new observations and shifting policy behavior.

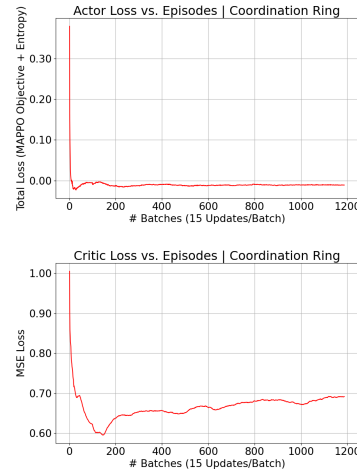


Figure 7: Loss curves for the actor (MAPPO Objective + Entropy, top) and critic (Mean-Squared Error, bottom) neural networks over all training batches, with 15 updates per batch in Coordination Ring during training.

3.2 Actor's Entropy Curve

Figure 8 illustrates how the actor's entropy changes as the number of policy updates increases. As mentioned, entropy controls the balance between exploration and exploitation during training. The actor's entropy begins at a high value but decreases rapidly in the early stages, then stabilizes at a lower value. This pattern is expected in agents that are learning effectively: they initially rely on random actions to explore the environment and occasionally discover optimal strategies. As training progresses, entropy decreases, indicating that the agents are making fewer random decisions and increasingly exploiting learned optimal behaviors. Unlike the actor's entropy curve in Cramped Room, which showed a steady decline throughout training, entropy in Coordination Ring decreases primarily during the first 200 batch updates before leveling off. This is likely due to the differing complexity of the two layouts; Coordination Ring is more complex and may require the agents to retain more stochasticity in their decision-making to remain adaptable to environmental demands.

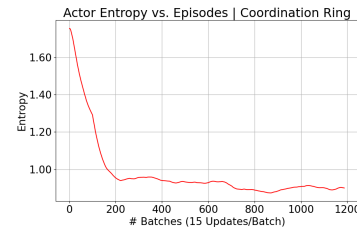


Figure 8: Entropy curve for the actor neural network over all training batches, with 15 updates per batch in Coordination Ring.

3.3 Performance During Training

Figure 9 illustrates the agents' performance during training, showing the number of dishes and number of soups delivered. As expected, and similar to their performance in Cramped Room, the agents improve at selecting more useful dishes and delivering soups during training. However, unlike in Cramped Room, the gap between the number of dishes picked up by each agent grows increasingly large. This is likely due to the more constrained layout, which limits movement and often forces one agent to step aside while the other performs key sub-tasks. The increased coordination demands also significantly impacted the average number of soups delivered during training: it took the agents 38,000 training episodes to reach an average of 7.0 deliveries per episode—about five times longer than in Cramped Room. This is not surprising, as MAPPO's centralized critic must learn to evaluate joint observations in a more complex environment, increasing sample complexity and thereby extending the training time.

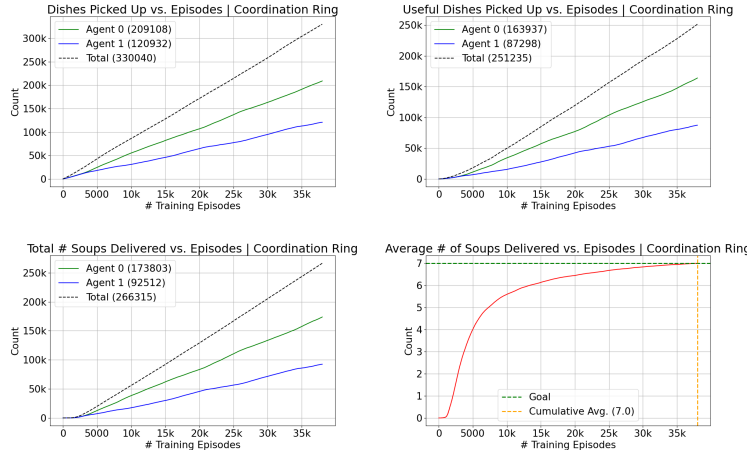


Figure 9: Plots showing the total (top left) and useful (top right) number of dishes picked up, as well as the total (bottom left) and average (bottom right) number of soups delivered by both agents during training in Coordination Ring.

3.4 Performance During Testing

Figure 10 illustrates how the agents performed in Coordination Ring using the policy that was learned during training. In terms of shaped rewards, the agents were able to close the gap between their earned rewards as training progressed. Although only one agent can occupy a key tile at a time, whether preparing ingredients or cooking and delivering soup, the balanced reward distribution suggests that the agents learned a strategy in which both perform sub-tasks simultaneously. The shaped reward pattern may also result from the use of decentralized execution, where each agent relies only on its own observations when acting in the environment. However, because the policy is trained using centralized information, including both agents' observations, the agents ultimately learn similar behaviors, leading to a more balanced distribution of rewards. In terms of the average number of soups delivered during testing, the agents started relatively low, improved rapidly, and eventually stabilized at a much higher level. The rapid increase may be due to the centralized critic, which helps reinforce effective strategies once the agents begin to discover useful behaviors (Albrecht et al. (2024)). After 100 testing episodes, the agents successfully solved the Coordination Ring layout, achieving an average of 7.8 soup deliveries per episode.

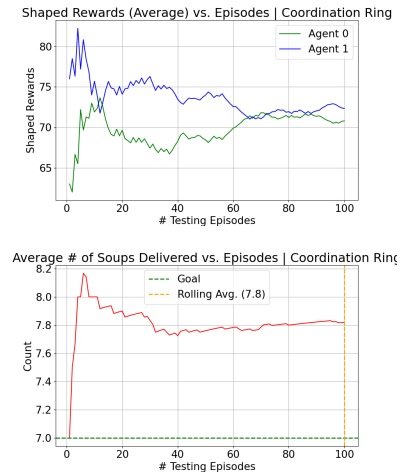


Figure 10: Average shaped rewards earned by each agent and average number of soups delivered by both agents during testing in Coordination Ring.

4 Counter Circuit

The final Overcooked layout tested was Counter Circuit, shown in Figure 11. All key tiles (onion, dish, pot, and delivery) are placed on opposite sides of the map, and the optimal strategy involves one agent passing ingredients across the counter while the other handles cooking and delivery (Carroll (2019)). Of the three layouts, Counter Circuit is the most difficult for the agents to learn, as success requires discovering an effective division of labor strategy, demanding a high level of coordination and timing.

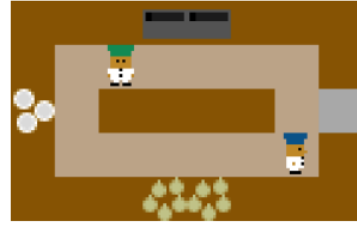


Figure 11: Counter Circuit layout with Agent 0 (green) and Agent 1 (blue)

4.1 Loss Curves for Actor and Critic

Figure 12 illustrates the actor’s and critic’s loss curves in Counter Circuit during training. Similar to the previous layouts, the actor’s loss curve in this layout rapidly decreases and stabilizes at a very low value. Although this layout is more complex than the others, MAPPO’s clipped policy updates and shared actor once again enable stable learning for both agents. Unlike the actor’s loss curve, the critic’s loss behaves unexpectedly, showing an increasing trend rather than the steady decrease observed in Cramped Room and Coordination Ring. This indicates that the critic is failing to generalize when predicting state values, likely due to the complexity of the agents’ joint observations and delayed reward signals. As a result, its predictions begin to diverge from the actual returns observed during training. This could be because the learning rate ($\alpha = 0.001$) is too high, resulting in instability, or that the batch size (32) is too small, leading to noisier gradient updates which may destabilize the critic (Restack (2025)).

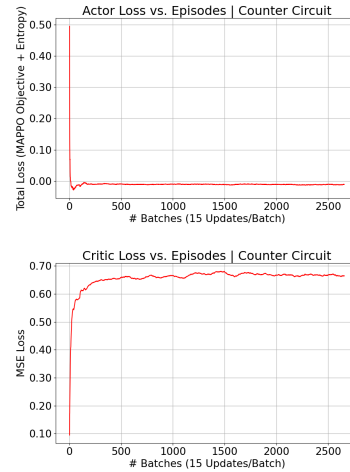


Figure 12: Loss curves for the actor (MAPPO Objective + Entropy, top) and critic (Mean-Squared Error, bottom) neural networks over all training batches, with 15 updates per batch in Counter Circuit during training.

4.2 Actor’s Entropy Curve

Figure 13 illustrates how the actor’s entropy changes as the number of policy updates increases. Similar to the other layouts, entropy decreases over the course of training in the Counter Circuit layout. This is expected, as a natural decline in entropy typically indicates that the agents are learning the environment and shifting from exploration to exploitation. However, unlike the other layouts, entropy in Counter Circuit shows more noticeable fluctuations even after convergence. This may be due to the entropy coefficient (0.01), which maintains a balance between random exploration and more deterministic decision-making (Restack (2025)), allowing the agents to stay adaptable in a complex environment. These fluctuations are expected given that the layout’s higher complexity may benefit from varying levels of randomness at different points in training, preventing the agents from getting stuck in local optima.

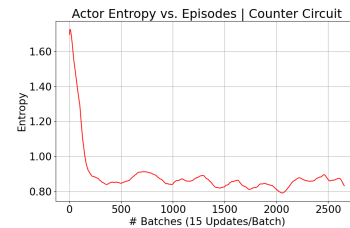


Figure 13: Entropy curve for the actor neural network over all training batches, with 15 updates per batch in Counter Circuit.

4.3 Performance During Training

Figure 14 illustrates the agents' performance during training. The plots show a stark difference in the number of dishes picked up and soups delivered by each agent. However, this does not imply that the agent with fewer pickups and deliveries learned poorly. Rather, it suggests that the agents successfully learned to divide responsibilities: Agent 1 primarily handled dish pickups and soup deliveries, while Agent 0 focused on complementary tasks that supported Agent 1's role. As mentioned, this division of labor is optimal for the Counter Circuit layout. It took the agents 85,000 training episodes to reach an average of 7.0 deliveries per episode—nearly three times longer than in Coordination Ring and more than 12 times longer than in Cramped Room. This extended training time is expected, given the significantly higher complexity and coordination demands of this layout. Additionally, the mismatch between the actor and critic loss curves, shown in Figure 12, may have further contributed to the prolonged training period.

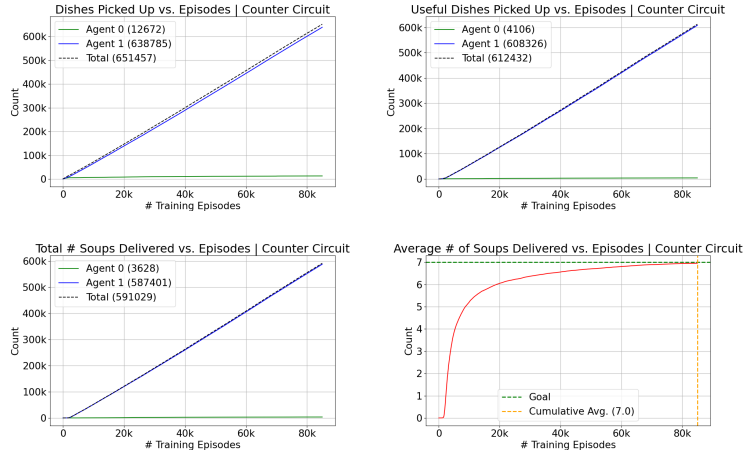


Figure 14: Plots showing the total (top left) and useful (top right) number of dishes picked up, as well as the total (bottom left) and average (bottom left) number of soups delivered by both agents during training in Counter Circuit.

4.4 Performance During Testing

Figure 15 illustrates how the agents performed in Counter Circuit using the policy that was learned during training. In terms of shaped rewards, there was an initial imbalance, which aligns with the agents adopting a division of labor. However, the rewards begin to converge around episode 30, suggesting that Agent 0 started contributing more directly to reward-generating actions, such as placing ingredients into the pot or occasionally delivering soup. This convergence in shaped rewards highlights that even without direct communication, the agents learned to implicitly coordinate their behaviors through shared training signals. The stability in later rewards suggests that the learned policy is consistent and well-formed, and that the agents are reliably performing their roles without major changes. In terms of the average number of soups delivered during testing, the agents started high, decreased, and then stabilized at a lower level. This initial drop may be due to residual randomness in the learned policy or slight differences between training and testing dynamics. The final average delivery rate surpassing the threshold confirms that, despite initial fluctuations, the learned policy generalized well to the testing environment. After 100 testing episodes, the agents successfully solved the Counter Circuit layout, achieving an average of 7.3 soup deliveries per episode.

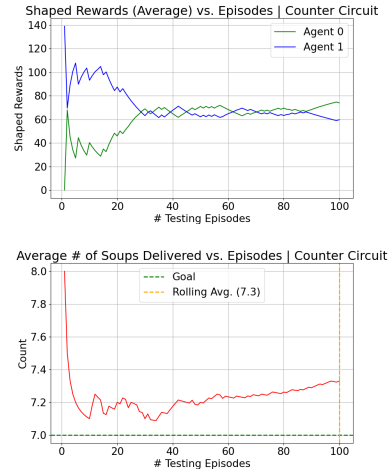


Figure 15: Average shaped rewards earned by each agent and average number of soups delivered by both agents during testing in Counter Circuit.

5 Conclusion

The purpose of this project was to evaluate the performance of reinforcement learning agents trained with Multi-Agent Proximal Policy Optimization (MAPPO), a policy gradient algorithm, in three layouts of the Overcooked environment.

In Overcooked, agents must learn to deliver as many soups as possible within a fixed time limit. MAPPO was selected as the training algorithm because it is well-suited for cooperative multi-agent tasks, aligning with the coordination and teamwork required in this environment. Furthermore, training was conducted using centralized training with decentralized execution (CTDE), where information from both agents is used to update the policy, but each agent acts using only its own local observations. This framework enables the agents to learn coordinated strategies during training while remaining independent during execution. This is particularly important in Overcooked, where agents must quickly adapt to each other's actions under strict time constraints.

The main objective for the agents was to achieve an average of more than seven soup deliveries per episode across three Overcooked layouts, each with progressively greater coordination complexity. The agents succeeded in all three, achieving average deliveries of 8.8, 7.8, and 7.3 per episode in Cramped Room, Coordination Ring, and Counter Circuit, respectively. These consistent successes highlight the effectiveness of MAPPO and CTDE in cooperative multi-agent settings, where stable policy learning is critical for enabling each agent to adapt to the other's actions and complete the objective within the time limit. Beyond achieving the delivery objective, MAPPO also supported stable learning, as demonstrated by the actor and critic loss curves for each layout, which generally followed smooth and predictable trends throughout training. CTDE further enabled the agents to divide labor and adopt complementary roles, particularly in the Counter Circuit layout. As shown in Section 4, Figure 14, Agent 1 completed the majority of deliveries. However, the shaped rewards in Section 4, Figure 15 shows that both agents earned similar rewards, indicating that Agent 0 was not idle, but instead performed meaningful sub-tasks to support the team. This coordination was possible because, during training, Agent 0 had access to Agent 1's observations, even though it relied only on its own observations during execution.

This project evaluated two hypotheses. First, MAPPO was expected to produce steadily increasing soup delivery performance during training due to its stability, which was confirmed by the results (Sections 2–4; Figures 4, 9, and 14). Second, the agents were expected to achieve the highest delivery rates in Cramped Room, followed by Coordination Ring, and the lowest in Counter Circuit, reflecting the increasing coordination complexity of the layouts. This was also confirmed by the results (Sections 2–4; Figures 5, 10, and 15). Overall, the findings demonstrate that MAPPO is effective for training multiple agents in Overcooked, as it allowed for stable learning and cooperative behaviors across layouts with varying coordination demands.

References

- Stefano V. Albrecht, Filippos Christianos, and Lukas Schäfer. *Multi-Agent Reinforcement Learning: Foundations and Modern Approaches*. MIT Press, 2024. URL <https://www.marl-book.com/download/marl-book.pdf>.
- et al. Carroll. On the utility of learning about humans for human-ai coordination. *arXiv preprint arXiv:1910.05789*, 2019. URL <https://arxiv.org/pdf/1910.05789>.
- OpenAI. Proximal policy optimization (ppo), 2025. URL <https://spinningup.openai.com/en/latest/algorithms/ppo.html>. Accessed: 2025-02-22.
- Restack. Ppo algorithm hyperparameters – hyperparameter tuning answer, 2025. URL <https://www.restack.io/p/hyperparameter-tuning-answer-ppo-algorithm-hyperparameters-cat-ai>. Accessed: 2025-02-22.